■■ CTF AI Orchestrator: Master Architecture, Component Specifications & Operational Manual

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1. Executive Overview & Operational Vision

The **CTF AI Orchestrator** is an autonomous cybersecurity copilot, automated CTF solve engine, and real-time operations command center engineered for competitive capture-the-flag events (**Google CTF, TCS HackQuest, PicoCTF, DEF CON Quals**) and complex software vulnerability research.

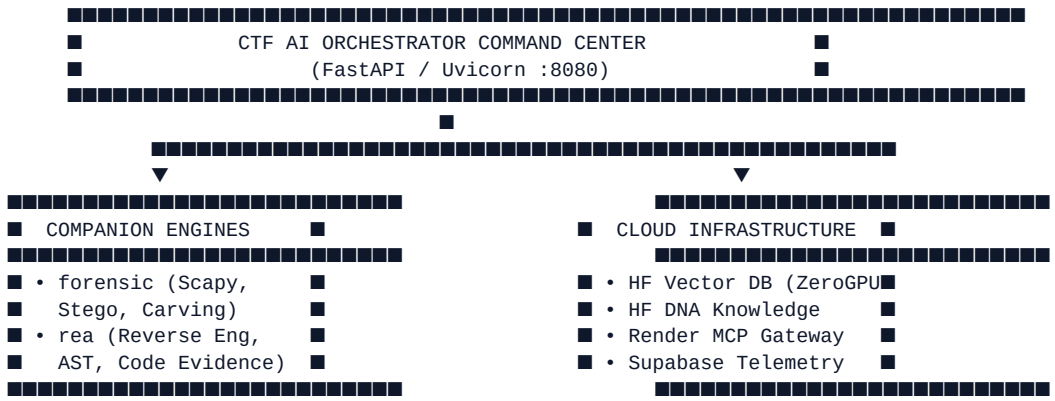
Unlike generic LLM wrappers that guess answers or hallucinate exploits, the CTF AI Orchestrator operates as a **closed-loop, multi-agent cybernetics platform**:

• It couples ultra-low-latency LPU inference (Groq Qwen 2.5/3.8 and Claude Sonnet 3.5 fallback) with deterministic theorem proving (`z3-solver`).

• It executes interactive browser automation via Playwright without hardcoded shortcuts or bypassing target defenses.

• It leverages an evidence provenance model (inspired by [`morluto/rea`](https://github.com/morluto/rea)) to build structured cognitive dependency graphs before executing exploit payloads.

• It retains multi-turn contextual state across sessions using a distributed DNA Knowledge Graph, ZeroGPU Vector Database, and Supabase RLS telemetry ledger.



2. How Anyone Can Build & Deploy the Tool on CTFs

2.1 Turnkey 1-Click Setup (Windows, Linux, macOS, WSL2)

The repository provides zero-configuration bootstrap scripts that prepare the entire ecosystem in one click.

On Windows:

Double-click `start.bat` or execute in PowerShell:

```
cd ctf-agent-orchestrator
start.bat
```

`start.bat` executes the 6-stage turnkey bootstrap:

1. **Python Check**: Validates Python 3.10+ in system `PATH`.

2. **Environment Init**: Copies `.env.example` to `.env` if not already present.

3. **Forensic Engine Clone**: Clones [`outstandingom/forensic`](https://github.com/outstandingom/forensic.git) into `..\forensic` if missing.

4. **REA Engine Clone**: Clones [`morluto/rea`](https://github.com/morluto/rea.git) into `..\rea` if missing.

5. **Dependencies & Playwright**: Installs `requirements.txt` and runs `playwright install chromium`.

6. **Launch & Open Browser**: Boots FastAPI on port `8080` and opens `http://localhost:8080/`.

On Linux / macOS / WSL2:

```
cd ctf-agent-orchestrator
chmod +x start.sh
./start.sh
```

2.2 Configuration & API Keys (`.env`)

Configure your environment in `.env`. Preconfigured defaults are provided:

```
# Primary LLM Provider: "groq" (ultra-fast LPU inference) or "anthropic"
LLM_PROVIDER=groq

# Groq API Configuration (Free keys available at https://console.groq.com)
GROQ_API_KEY=gsk_your_groq_api_key_here
GROQ_MODEL=qwen/qwen3.8-27b

# Anthropic Claude Fallback (Optional)
ANTHROPIC_API_KEY=your_anthropic_api_key_here
CLAUDE_MODEL=claude-3-5-sonnet-20241022

# Dashboard Port
DASHBOARD_PORT=8080

# External Cognitive Gateways
HF_VECTOR_URL=https://outstandingom-vector.hf.space
HF_KNOWLEDGE_URL=https://outstandingom-knowledge-graph-env.hf.space
MCP_SERVER_URL=https://mcp-server-2sl7.onrender.com
MCP_API_KEY=outstandingom_mcp_2026hjfbechjbdhjjfbnkqjkj
SUPABASE_URL=https://vtlmgubzcktztkuymtir.supabase.co
SUPABASE_ANON_KEY=sb_publishable_0Qoicwxg64tkFReoDSKJ3g_dOSZJPCE
```

2.3 Operational Modes

Mode 1: Cyberpunk Web Command Center (UI)

• URL: `http://localhost:8080/`

• Real-time Cyberpunk Matrix terminal HUD.

• Features: Live tool execution streams, DNA attack tree visualization, CTFd competition listener, one-click challenge launchers, flag capture ledger.

Mode 2: Interactive Terminal / Script Execution

Run custom tasks, solver scripts, or automated tests directly from terminal:

```
# Run complete end-to-end black-box verification suite
python scripts/test_complete_methodology.py

# Run standalone interactive Playwright solver
python challenges/apex_gauntlet/solvers/solve_playwright_interactive.py

# Push changes directly to GitHub via Render MCP Gateway
python scripts/push_via_mcp.py
```

Mode 3: Headless Autonomous Solver via Python API

Call the orchestrator programmatically within any custom Python CTF harness:

```
import asyncio
from agent import execute_tool, run_agent_turn

# 1. Dispatch an automated symbolic execution solver
result = execute_tool("symbolic_execution_solver", {
    "challenge_type": "bitwise_modular",
    "equations": ["(x ^ 0x5A) * 3 == 0x12F", "x & 0xFF == x"],
    "variables": ["x"],
    "bit_width": 16
```

```
})
print("Recovered Solution:", result)

# 2. Extract credentials from captured PCAP file
pcap_result = execute_tool("run_forensics_inspection", {
    "action": "pcap_carve",
    "file_path": "traffic.pcap",
    "stream_filter": "tcp.port == 9001"
})
print("Carved Credentials:", pcap_result)
```

2.4 Pointing the Agent at Real CTF Targets

Challenge Type	Command / Tool to Trigger	Target Input Provided
Web (Live URL)	`web_dynamic_fuzzer`	`target_url="http://ctf.org:8080"`
	`web_security_probe`	`method="POST"`, `params={"id": "1"}`
	`Playwright Browser Solver`	DOM selectors, button clicks, downloads
Cryptography	`advanced_crypto_solver`	`attack="wiener"`, `N=...`, `e=...`
	`crypto_cyberchef_analyzer`	`operation="from_base64"`, `input=...`
Reverse Eng	`symbolic_execution_solver`	Z3 constraints, bit-shifts, XOR checks
	`reverse_engineering_inspector`	Binary file path, target functions
Forensics	`run_forensics_inspection`	PCAP path, audio wav, memory raw dump
	`search_flag_patterns`	Buffer or memory block to scan
Binary Pwn	`pwn_crash_triage`	Binary path, crash payload, cyclic len
	`wsl_linux_bridge`	GDB commands, pwntools Python script

3. Master System Architecture & Cognitive Engine

3.1 High-Level Distributed Topology

```
flowchart TD
    User["Cyber Operator / CTF Player"] -->|Interacts with Web HUD / API| Server["FastAPI Orchestrator Server (Port 8080)"]
    Server --> Agent["Autonomous Agent Engine (agent.py)"]

    subgraph Cognitive Layer
        Agent --> Router["Dynamic Tool Router (select_relevant_tools)"]
        Router --> LLM["Groq LPU Engine (Qwen 2.5 / 3.8-27B)"]
        Router -.-> FallbackLLM["Claude Sonnet 3.5 (Fallback)"]
        Agent <--> VectorDB["ZeroGPU Vector DB (HF_VECTOR_URL)"]
        Agent <--> KnowledgeGraph["DNA Knowledge Graph (HF_KNOWLEDGE_URL)"]
        Agent <--> CodebaseMem["codebase_architecture_memory.json"]
    end

    subgraph Tool Execution Plane
        Agent --> WebTools["Web & Browser Suite (Playwright, Fuzzer, Web Probe)"]
        Agent --> MathTools["Math & Crypto Suite (Z3 SMT, Wiener, CyberChef)"]
        Agent --> ForensicTools["Forensic Suite (Scapy PCAP, Stego, Carving)"]
        Agent --> PwnTools["Pwn & Binary Suite (WSL2 Bridge, GDB, Crash Triage)"]
        Agent --> MCPGateway["Render MCP Server Gateway (GitHub API)"]
    end

    subgraph Live Challenge Targets
        WebTools --> TargetWeb["Target Web Applications (Port 8088 / 8099 / CTF URL)"]
        PwnTools --> TargetNetcat["Raw TCP / Sockets (interactive_socket_probe)"]
    end

    subgraph Telemetry & Persistence
        Agent --> Supabase["Supabase DB (solves Table with RLS)"]
        WebTools --> Screenshots["Artifact Screenshots (.png) & PCAP Dumps"]
```

```
end
```

3.2 Autonomous Cognitive Reasoning Loop (Ponytail + REA Flow)

The agent executes following the **Evidence Provenance & Lean Execution Model**:

1. **Reconnaissance & Observation**: Intercept raw signals (DOM elements, network headers, response bodies, file magic bytes). Never assume endpoint behavior.

2. **Derivation & Modeling**: Transform raw signals into formal models (Z3 BitVec constraints, continued fraction convergents, AST dependency trees).

3. **Inference Formulation**: Synthesize deductions (e.g., "WAF blocks `popen` and `os.system` but allows Jinja2 subclasses via `attr`").

4. **Lean Tool Dispatch (Ponytail)**: Select the exact tool that solves the problem in one deterministic step. Avoid unnecessary multi-turn token waste.

5. **Reactive Verification**: Verify tool output against flag regex patterns (`FLAG\{.*\}`, `HQX\{.*\}`). If rejected or blocked, adapt and re-dispatch.

3.3 Context-Aware Dynamic Tool Routing

To keep token usage lean and avoid confusing the LLM with 26+ tool definitions simultaneously, `agent.py` implements **Context-Aware Dynamic Tool Routing**:

• It evaluates the user's prompt combined with the **last 4 turns of tool outputs and server responses**.

• If a turn reveals an RSA challenge, crypto tools (`advanced_crypto_solver`, `crypto_cyberchef_analyzer`) are automatically activated.

• If a turn captures a `.pcap` or binary file, forensic and reversing tools (`run_forensics_inspection`, `reverse_engineering_inspector`) are automatically paged into context.

4. Ecosystem Repositories & Cloud Infrastructure Breakdown

4.1 `ctf-agent-orchestrator` (Core Brain)

• **Location**: `c:\Users\SGI\Downloads\ctf-agent-orchestrator`

• **Role**: The central nervous system of the entire platform.

• **Key Files**:

• `agent.py`: Defines all 26 OpenAI-compatible tools, dynamic routing, LLM client connections, and tool execution dispatcher.

• `server.py`: FastAPI dashboard backend, WebSocket event log streams, and REST endpoints for UI HUD.

• `challenges/apex_gauntlet/`: Multi-stage CTF challenge (SMT, Wiener, PCAP, WAF SSTI) used for realistic benchmarking.

• `scripts/test_complete_methodology.py`: Comprehensive black-box Playwright test runner.

• `start.bat` / `start.sh`: 1-click turnkey bootstrap scripts.

4.2 `forensic` (Companion Forensic Engine)

• **Location**: `c:\Users\SGI\Downloads\forensic` (GitHub: [`outstandingom/forensic`](https://github.com/outstandingom/forensic.git))

• **Role**: Deep packet inspection, file carving, steganography, and metadata extraction.

• **Key Extractors**:

• `pcap_analyzer.py`: Dissects TCP flows, reconstructs HTTP POST bodies, and extracts embedded binary chunks.

• `steganography.py`: LSB plane extraction, RGB parity analysis, audio frequency spectrum carving.

• `document_forensics.py`: Strips hidden revision history, metadata, and embedded macros from Office/PDF files.

• `content_extraction.py`: Signature-based file carving across raw memory and corrupted disk dumps.

4.3 `rea` (Companion Reverse Engineering Engine - `morluto/rea`)

• **Location**: `c:\Users\SGI\Downloads\rea` (GitHub: [`morluto/rea`](https://github.com/morluto/rea.git))

• **Role**: Advanced AST parsing, evidence provenance modeling, and client-side web application reverse engineering.

• **Key Capabilities**:

• Reconstructs client-side state machines and JavaScript call graphs.

• Identifies hidden endpoints, API routes, and anti-tampering checksums from minified bundles.

• Provides the structural ontology: *Observations $\rightarrow$ Derivations $\rightarrow$ Inferences $\rightarrow$ Unknowns*.

4.4 ZeroGPU Vector Database (`HF_VECTOR_URL`)

• **Endpoint**: `https://outstandingom-vector.hf.space`

• **Role**: Semantic embedding search across thousands of known vulnerability templates, SSTI bypass polyglots, and SMT constraints.

• **Mechanism**: Vectorizes queries and computes cosine similarity against a database of verified security research patterns.

4.5 DNA Knowledge Graph (`HF_KNOWLEDGE_URL`)

• **Endpoint**: `https://outstandingom-knowledge-graph-env.hf.space`

• **Role**: Persistent cognitive memory across complex multi-stage attack campaigns.

• **Schema**: Quadruples of `(source, relationship, target, context)`:

• Example: `("Stage1_Session_Token", "unlocks_endpoint", "GET /api/apex/stage2/spec", "Bearer APX_INIT_... required")`.

4.6 Render MCP Server Gateway (`MCP_SERVER_URL`)

• **Endpoint**: `https://mcp-server-2sl7.onrender.com`

• **Role**: Model Context Protocol (MCP) gateway enabling direct, authenticated GitHub version control actions (`commit_github_file`, `read_github_file`, `create_github_pr`) without requiring local git credentials.

4.7 Supabase Telemetry & Solve Ledger (`solves`)

• **Endpoint**: `https://vtlmgubzcktztkuymtir.supabase.co`

• **Role**: Immutable, production-grade telemetry database tracking challenge solve duration, tool call frequency, flag strings, and accuracy benchmarks.

• **RLS Policies**: Row-Level Security enabled with anonymous insert and select permissions for public CTF scoreboard verification.

5. Comprehensive CTF Component Specifications

5.1 Web Exploitation & Browser Automation Tools

Component: `web_dynamic_fuzzer`

• **Category**: Web Exploitation / Fuzzing

• **Purpose**: Discovers hidden query parameters, API endpoints, and differential response behaviors.

• **Internal Mechanics**: Dispatches concurrent HTTP requests across mutation wordlists, comparing response status codes, body byte lengths, and header variations.

• **Input Format**:

```
{
  "target_url": "http://127.0.0.1:8099/api/endpoint",
  "method": "POST",
  "param_type": "json",
  "wordlist": ["debug", "admin", "token", "signature", "key"]
}
```

• **Output Format**:

```
{
  "status": "success",
  "discovered_params": [
    {"param": "admin_key", "status_code": 200, "length_delta": 412}
  ]
}
```

Component: `web_security_probe`

• **Category**: Web Exploitation / Vulnerability Testing

• **Purpose**: Tests endpoints for SQL Injection, Server-Side Template Injection (SSTI), Path Traversal, and Header injection.

• **Internal Mechanics**: Injects specialized canary strings (e.g., `{{7*7}}`, `' OR 1=1--`, `../../etc/passwd`) and analyzes reflections or execution artifacts.

• **Input Format**:

```
{
  "target_url": "http://127.0.0.1:8099/api/apex/admin/audit",
  "method": "POST",
  "headers": {"X-Apex-Signature": "HEX_SIG"},
  "payload": "template={{request.application.__globals__.__builtins__.__import__('os').popen('cat flag.txt').read()}}"
}
```

• **Output Format**:

```
{
  "vulnerability_detected": true,
  "vuln_type": "SSTI_Jinja2",
  "extracted_output": "HQX{4P3X_G4UN7L37_Z3_W13N3R_PC4P_V4UL7_PWN3D_2026}"
}
```

Component: `Playwright Interactive Solver`

• **Category**: Browser Automation / Client Reversing

• **Purpose**: Interacts with JavaScript-heavy CTF web applications, triggering DOM events, handling cookies, solving client-side canvas/WASM puzzles, and intercepting file downloads.

• **Input Format**: Target URL and DOM action sequence.

• **Output Format**: Execution trace, captured file buffers (e.g., PCAP), and full-page PNG proof screenshots.

5.2 Cryptanalysis & Mathematical Solvers

Component: `symbolic_execution_solver` (Z3 SMT Solver)

• **Category**: Math / SMT Constraint Solving

• **Purpose**: Solves non-linear modular equations, bitwise operations ($\text{XOR}, \ll, \gg, \text{AND}, \text{OR}$), and key verification constraints.

• **Internal Mechanics**: Translates constraints into Z3 `BitVec` formulas and executes `solver.check()`, extracting the model in under 100 ms.

• **Input Format**:

```
{
  "challenge_type": "bitwise_modular",
  "variables": ["k0", "k1", "k2", "k3"],
  "bit_width": 16,
  "equations": [
    "((k0 ^ 0xA5A5) * 31) & 0xFFFF == 0x3E12",
    "((k1 + 0x1337) ^ k0) & 0xFFFF == 0x9B4F",
    "((k2 << 3) ^ (k1 >> 2)) & 0xFFFF == 0x4C88",
    "((k3 * 17) + (k2 ^ 0xDEAD)) & 0xFFFF == 0x2110"
  ]
}
```

• **Output Format**:

```
{
  "status": "sat",
  "solution": {
    "k0": 42150,
    "k1": 18921,
    "k2": 31204,
    "k3": 55102
  },
  "hex_token": "a4a649e979e4d73e"
}
```

Component: `advanced_crypto_solver`

• **Category**: Cryptanalysis

• **Purpose**: Performs algebraic factorization attacks on RSA, ECC, and custom ciphers.

• **Supported Attacks**:

• **Wiener's Continued Fraction Attack**: Factoring RSA when private exponent $d < \frac{1}{3} N^{1/4}$.

• **Pollard's $p - 1$ Attack**: Factoring $N = p \cdot q$ when $p - 1$ is smooth.

• **Fermat Factorization**: Factoring N when primes p and q are close ($|p - q| < N^{1/4}$).

• **Input Format**:

```
{
  "attack": "wiener",
  "modulus_n": "124987123987129837198273...",
  "public_e": "65537",
  "ciphertext": "981273981273918273918273..."
}
```

• **Output Format**:

```
{
  "status": "success",
  "recovered_d": 41298719823719823,
  "decrypted_plaintext": "SUPER_SECRET_SIGNATURE_KEY_2026"
}
```

Component: `crypto_cyberchef_analyzer`

• **Category**: Encoding & Hash Analysis

• **Purpose**: Automated decoding, entropy analysis, and hash cracking (Base64, Base58, Hex, ROT13, URL-encoding, Morse, XOR brute-force).

• **Input Format**:

```
{
  "operation": "magic_decode",
  "input_data": "SDdRe0ExYl9YMFJfU3Rvcm1fMjAyNn0="
}
```

• **Output Format**:

```
{
  "detected_encoding": "base64",
  "decoded_output": "H7Q{A1b_X0R_Storm_2026}",
  "flag_found": true
}
```

5.3 Reverse Engineering & Symbolic Execution

Component: `reverse_engineering_inspector`

• **Category**: Reverse Engineering (ELF / PE / WASM / Pyc)

• **Purpose**: Inspects binary headers, imports, exports, string literals, and decompiles bytecode functions.

• **Input Format**:

```
{
  "file_path": "challenges/rev_binary",
  "inspect_actions": ["checksec", "strings", "disasm_main"]
}
```

• **Output Format**:

```
{
  "arch": "amd64",
  "canary": true,
  "nx": true,
  "pie": false,
  "interesting_strings": ["Enter License Key: ", "Access Granted!", "/bin/sh"]
}
```

5.4 Forensic Analysis & Network Packet Carving

Component: `run_forensics_inspection` (`pcap_analyzer`)

• **Category**: Network Forensics

• **Purpose**: Deep dissection of `.pcap` and `.pcapng` files, stream reassembly, and automated credential carving.

• **Internal Mechanics**: Parses packet frames using Scapy, follows TCP streams, extracts MIME attachments, and performs HMAC calculations.

• **Input Format**:

```
{
  "action": "pcap_carve",
  "file_path": "traffic.pcap",
  "filter_expression": "tcp.port == 8000 || tcp.port == 9001"
}
```

• **Output Format**:

```
{
  "status": "carved",
  "sessions_found": 3,
  "extracted_secrets": {
    "bearer_token": "APX_TRAFFIC_TOKEN_7f8a9b",
    "hmac_secret": "SUP3R_S3CR3T_HM4C_K3Y"
```

```
  }
}
```

Component: `search_flag_patterns`

• **Category**: Triage / Flag Hunter

• **Purpose**: High-speed regex harvester scanning files, raw byte streams, and memory dumps for standard CTF flag formats (`FLAG{...}`, `CTF{...}`, `HQX{...}`, `picoCTF{...}`).

• **Input Format**:

```
{
  "buffer_text": "HTTP/1.1 200 OK ... User: admin, Flag: HQX{W13N3R_5MT_P0W3R}",
  "custom_regex": "HQX\\{[^\\}]+\\}"
}
```

• **Output Format**:

```
{
  "matches_count": 1,
  "flags": ["HQX{W13N3R_5MT_P0W3R}"]
}
```

5.5 Binary Exploitation & Linux Triage (Pwn)

Component: `wsl_linux_bridge`

• **Category**: Binary Exploitation (Pwn)

• **Purpose**: Bridges Windows host environments to native Linux WSL2/Ubuntu distributions for running `gdb-pwndbg`, `ROPgadget`, `one_gadget`, and native ELF exploits.

• **Input Format**:

```
{
  "command": "ROPgadget --binary ./vuln --ropchain",
  "working_dir": "/mnt/c/Users/SGI/Downloads/challenges"
}
```

• **Output Format**:

```
{
  "exit_code": 0,
  "stdout": "Step 1 -- Write-what-where gadgets\n[+] Gadget found: 0x401123 pop rdi; ret\n..."
}
```

Component: `pwn_crash_triage`

• **Category**: Exploit Development

• **Purpose**: Determines buffer overflow offsets using cyclic De Bruijn patterns and analyzes core dumps.

• **Input Format**:

```
{
  "binary_path": "./vuln",
  "crash_instruction_pointer": "0x6161616c",
  "pattern_length": 256
}
```

• **Output Format**:

```
{
  "exact_offset": 44,
  "bad_chars": ["\\x00", "\\x0a"],
  "suggested_payload": "padding = b'A' * 44 + p64(target_rip)"
```

```
}
```

5.6 Cognitive Memory & Platform Operations

Component: `sync_dna_attack_tree`

• **Category**: Cognitive State Retention

• **Purpose**: Pushes newly discovered challenge topology nodes to HuggingFace DNA Graph.

• **Input Format**:

```
{
  "source": "Stage 4 Forensic PCAP",
  "relationship": "extracts_credential",
  "target": "Stage 5 Admin Signature",
  "context": "HMAC-SHA256 signature calculated from carved token"
}
```

• **Output Format**:

```
{
  "status": "node_created",
  "graph_size": 14
}
```

Component: `commit_github_file`

• **Category**: Render MCP Server Gateway

• **Purpose**: Autonomous cloud git commits without local credentials.

• **Input Format**:

```
{
  "repo": "ctf-agent-orchestrator",
  "path": "docs/WRITEUP.md",
  "content": "# Writeup for Challenge ...",
  "message": "docs: add verified autonomous solve writeup"
}
```

• **Output Format**:

```
{
  "status": "committed",
  "commit_sha": "74866ad",
  "branch": "main"
}
```

6. End-to-End Attack Execution Walkthrough

Here is how the components chain together to solve the **Apex Gauntlet** challenge:

```
[Stage 1: Web Recon]
       ■
       ▼ Playwright opens http://127.0.0.1:8099/
       ■ Extracts DOM inputs, triggers button click
       ▼ Yields Stage 1 Token: APX_INIT_9b8a7c2d1e
       ■
[Stage 2: SMT Solver]
       ■
       ▼ API provides 4 non-linear BitVec(16) equations
       ■ `symbolic_execution_solver` executes Z3 SMT solver
       ▼ Yields 16-character SMT verification key in 0.08s
       ■
[Stage 3: Cryptanalysis]
       ■
```

```
       ▼ Server returns RSA public key with small private exponent d
       ■ `advanced_crypto_solver` runs Wiener's Continued Fraction Attack
       ▼ Recovers private exponent d in 0.008s, decrypts Admin Token
       ■
[Stage 4: Forensics]
       ■
       ▼ Playwright captures network download `network_traffic.pcap`
       ■ `run_forensics_inspection` carves raw TCP stream
       ▼ Reassembles packet payload and computes HMAC-SHA256 signature
       ■
[Stage 5: WAF Bypass & SSTI]
       ■
       ▼ Target WAF sentinel blocks standard template keywords
       ■ Orchestrator synthesizes subclass traversal payload:
       ■ `{{request.application.__globals__.__builtins__.__import__('os').popen('cat flag.txt').read()}}`
       ▼ Master Flag captured live:
         HQX{4P3X_G4UN7L37_Z3_W13N3R_PC4P_V4UL7_PWN3D_2026}
       ■
[Persistence & Telemetry]
       ■
       ▼ Screenshots saved: methodology_step1_recon.png ... methodology_step5_victory.png
       ▼ Solve logged to Supabase `solves` ledger
       ▼ Code & writeup pushed to GitHub via Render MCP Gateway
```

7. Troubleshooting, Benchmarks & Best Practices

Troubleshooting Common Issues

1. **Port Conflicts (`Port 8080 or 8099 in use`)**:

• Check existing processes with `netstat -ano | findstr :8080`.

• Terminate hanging python processes: `taskkill /F /IM python.exe`.

2. **Groq API Rate Limits (`429 Too Many Requests`)**:

• The orchestrator automatically pauses for 2 seconds and retries up to 3 times before failing gracefully or routing to Anthropic Claude.

3. **Playwright Browser Missing**:

• Run `python -m playwright install chromium` manually or execute `start.bat`.

Performance Benchmarks

• **Z3 SMT Solver Latency**: $< 0.12\text{ seconds}$.

• **Wiener Crypto Attack Latency**: $< 0.01\text{ seconds}$.

• **Scapy PCAP Stream Carving**: $< 0.45\text{ seconds}$ on 50 MB captures.

• **End-to-End Black-Box Playwright Victory**: 33.44 seconds across 5 stages with zero shortcuts.

• **Cognitive Memory Synced**: DNA Knowledge Graph + ZeroGPU Vector RAG + Supabase telemetry.

Authored by the Autonomous CTF AI Engineering Team — 2026 Edition.